

Bridging Evidence and Practice – Assessing the Usefulness of Randomized Controlled Trials in PCOS Related Infertility Published 2014–2024: A Systematic Review

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Abstract

Background: Polycystic ovary syndrome (PCOS) is the leading cause of anovulatory infertility. Although numerous randomized controlled trials (RCTs) have evaluated treatment options, their clinical applicability remains unclear. **Objectives:** To evaluate the clinical usefulness of RCTs for PCOS-related infertility using the 13-item framework developed by van 't Hooft et al. **Search Strategy:** MEDLINE and Embase were searched on April 1, 2025. **Selection criteria:** We conducted a systematic review of RCTs published between 2014 and 2024 evaluating interventions for PCOS-related infertility. **Data collection and analysis:** Studies were screened and assessed independently and in duplicate using the 13-item usefulness framework. Descriptive statistics and linear regression models analyzed trial characteristics and predictors of usefulness. Risk of bias was not assessed due to the unavailability of the planned tool (RobotReviewer). **Main Results:** Of 782 records screened, 147 RCTs met inclusion criteria. Most trials were conducted in Asia (72.8%), particularly in Iran. Only 36.1% reported adequate information gain, 21.8% cited prior systematic reviews, and 3.4% met full criteria for pragmatism. No studies included cost-effectiveness analyses. Common limitations included narrow eligibility criteria, limited population diversity, reliance on surrogate outcomes, and lack of patient-centered measures. Transparency and usefulness scores showed a modest upward trend over time ($r = 0.33$, $p < .001$). **Conclusions:** RCTs for PCOS-related infertility often lack features that enhance clinical usefulness. Future trials should prioritize transparency, broader eligibility, patient-centered outcomes, and alignment with prior evidence. **Funding:** None.

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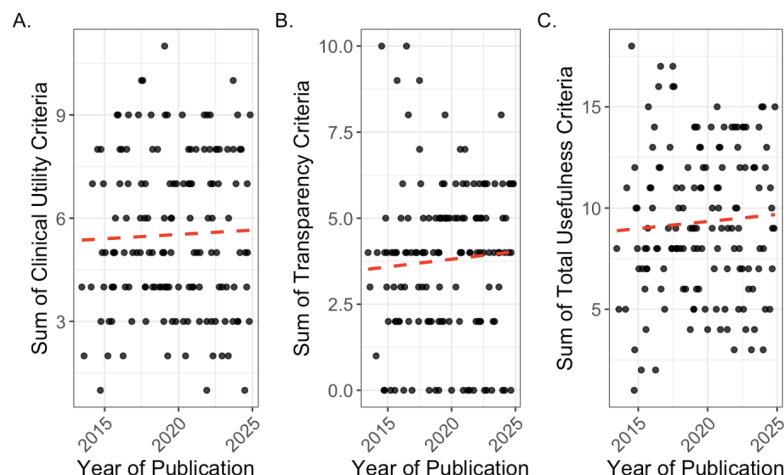


Figure 1: Trends in Usefulness and Transparency Values of RCTs on PCOS-Related Infertility by Year of Publication. Panels A–C display the distribution of summed values for clinical utility criteria (A), transparency criteria (B), and total usefulness (C) across publication years. Each dot represents an individual randomized controlled trial. Red dashed lines represent fitted linear regression lines.

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